

Gradient Boosting Recovers Fill Probability Signal That a Deep Survival Model Misses

A false negative about data content, from 70 days of
cryptocurrency limit order book data

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Abstract

Retail and non-institutional algorithmic traders operating under execution delays of 200 ms to 2s face substantial stale-quote risk: the book can move before a quote reaches the exchange. A natural response is to substitute prediction for speed, estimating the probability that a quote fills given current limit order book (LOB) state. Whether this is possible at the data frequency available to such traders is an empirical question, and we find that whether the available signal is recovered depends strongly on model class.

Using 269,330 LOB snapshots collected continuously over 70 days from Kraken across BTC/USD, ETH/USD and SOL/USD, we benchmark fill probability estimators against a deliberately trivial reference: a 32-cell lookup table holding only the mean historical fill rate for each combination of quote offset, latency window and side. Under a snapshot-grouped temporal split, gradient-boosted trees over microstructure features beat this reference on all three assets, by +0.174, +0.097 and +0.045 AUC (mean +0.105). An LSTM survival model over the same features, given a 50-snapshot temporal window the trees never receive, beats it on none, scoring -0.012, -0.007 and -0.116 against the table.

The positive result survives our leakage and specification checks. It holds under a split that prevents any snapshot appearing in both training and test data, and it strengthens when all price-derived columns are removed, ruling out the trees partitioning on price level as a proxy for time period. A boosted model restricted to the three conditioning variables reproduces the lookup table to four decimal places, confirming the comparison is set up correctly.

We also report a conditioning defect that invalidated our own earlier results, in which quote offset never reached the model, and which we retain as a reproduction control. Corrected and uncorrected forms differ by 0.15 C-index, and the uncorrected form performs at chance across five seeds.

The practical implication reverses the framing we began with. Microstructure signal is present at roughly 90 observed LOB updates per hour and is worth about 0.10 AUC over quote geometry alone. Extracting it, in our experiments, did not require more data, finer data or a larger model; it required a model class suited to tabular features. Our tested deep survival architecture, adapted from the higher-frequency literature, did not recover it.

Keywords: market making, market microstructure, limit order book, fill probability, execution latency, gradient boosting, model selection

JEL: G12, G14, C45, C61, G23

1 Introduction

In competitive markets, speed is capital. Firms dominating modern equity and cryptocurrency liquidity provision have invested heavily in compressing execution latency to microseconds. Microseconds do not matter to the real economy; being first to respond to a price signal is profitable, repeatedly, at scale. This raises a precise question: is speed itself the valuable resource, or is it a proxy for the ability to act on information before it becomes stale?

If the latter, speed might not be strictly necessary. A participant who cannot act in microseconds could still act correctly, provided they can predict what the market will look like when their order arrives. Earlier versions of this work proposed *latency-compensating market making* on that basis, in which models trained on LOB microstructure substitute predictive accuracy for execution speed.

This version reports the empirical status of the predictive component. The signal exists. Our original model could not find it.

1.1 Two findings

Microstructure predicts fills at retail data frequency. Gradient-boosted trees over microstructure features beat a lookup table on quote offset, latency and side by +0.174, +0.097 and +0.045 AUC on BTC, SOL and ETH. The effect appears on every asset, survives a snapshot-grouped split, and strengthens when price-derived features are removed.

A deep survival model extracts none of it. An LSTM over the same features, with a 50-snapshot temporal window the trees never see, scores at or below the lookup table on all three assets. Pooled, it matches the table to within 0.0003 AUC.

The gap between these findings is the paper. Both models had the same features, the same labels and the same split. One recovered roughly 0.10 AUC of signal and the other recovered none.

1.2 How we arrived here

We report the sequence because the intermediate conclusions were wrong in instructive ways.

Version 1 presented results on synthetic LOB data. Those results were encouraging and uninformative: a geometric Brownian motion price process with a hand-specified fill rule contains whatever structure the specification puts there.

On real data our first results fell below chance. A diagnostic located a defect in the model’s input pipeline (Section 6). Correcting it produced a real improvement, and we then compared the corrected model against a lookup table, a comparison that should have come first. The model did not beat it. We concluded microstructure carried no signal at this data frequency and wrote that conclusion up.

Running a gradient-boosted baseline, which that draft had itself flagged as missing, overturned the conclusion within twenty minutes. The features were informative throughout. The measurement instrument was not.

We draw the methodological point out in Section 9. Inferring a data limitation from one model’s failure is a specific and avoidable error, and we made it twice, for two different underlying reasons.

1.3 Contributions

1. We show microstructure features predict our replay-defined fill outcome at roughly 90 observed LOB updates per hour, worth a mean +0.105 AUC over quote geometry, on three assets (Section 7).

2. We show an LSTM survival model of the kind adapted from higher-frequency work extracts none of this signal, despite receiving strictly more information than the model that does (Section 7.1).
3. We release 269,330 timestamped 10-level LOB snapshots with the full collection, feature and evaluation pipeline (Section 4).
4. We report a leakage audit for the positive result: grouped splits, price removal, a shuffled-split contrast and a control fit that reproduces the reference (Section 8).
5. We document a conditioning defect that invalidated our own prior results and retain it as a reproduction control (Section 6).

1.4 Scope

Because the interpretive distance between what we measured and what the result is likely to be read as saying is large, we state the boundaries before the evidence.

Established. In our Kraken sample, covering three cryptocurrency pairs from June to August 2026 at roughly 90 observed LOB updates per hour, microstructure features carry predictive content for our replay-defined fill outcome beyond quote offset, latency and side, recoverable by gradient-boosted trees.

Not established: economic exploitability. AUC of 0.78 is not a trading strategy. We run no backtest, no transaction cost analysis and no live evaluation. The distance between a predictive edge and a profitable one is where most of the difficulty in this problem lies, and we do not cross it.

Not established: realised fills. Labels are reconstructed from subsequent book states and do not model queue position, so the measured outcome is an optimistic proxy for a late-arriving limit order. We show discrimination for a replay-defined fill, not for a realised execution.

Not established: calibration. AUC measures ranking. It does not establish that a predicted 0.70 corresponds to a 70% fill rate. This matters because the abstention rule in (5) thresholds a probability, and a well-ranked but poorly calibrated estimator would not support it without recalibration.

Not established: other deep architectures. We tested one architecture at one hyperparameter setting. A fair reading is that our implementation, hyperparameters or survival objective are at fault rather than deep learning generally.

Not established: generality. One venue, one 70-day window, three cryptocurrency pairs, one feed type. We make no claim about equity venues, other sampling frequencies or other periods.

Not tested. The reinforcement learning agent proposed in Version 1 consumed estimates from the defective model and was trained only on synthetic data. We report no RL results (Section 7.4).

2 Related Work

2.1 Market making and latency

The modern framework originates with Garman [1976] and Ho and Stoll [1981], formalised as a stochastic control problem by Avellaneda and Stoikov [2008]. Extensions incorporate adverse selection and execution uncertainty [Cartea et al., 2018], and Cartea and Sánchez-Betancourt [2023] introduce stochastic delay into optimal execution.

Nearly all analytical market making models assume negligible execution latency. Lehalle and Mounjid [2022] show the assumption is load-bearing: the value of predictive signals about future

order flow is systematically eroded by latency. Our results bear on the premise of that argument, since they establish a usable signal exists at retail data frequency in the first place.

2.2 The latency arms race

Moallemi and Saglam [2013] give the first formal model of latency cost. Budish et al. [2015] characterise continuous-time limit order books as generating structurally wasteful speed investment. Aquilina et al. [2022] quantify this, finding latency arbitrage races roughly once per minute per symbol in FTSE 100 stocks with modal durations of 5–10 microseconds, and estimating that eliminating them would cut effective spreads by about 17%.

Proposed remedies (the IEX speed bump of Hu, 2019, frequent batch auctions in Budish et al., 2015, micro-burst fees in Brolley and Zoican, 2023) operate at the exchange level and require regulatory or venue adoption. None offers recourse to an individual trader on existing infrastructure.

2.3 Fill probability estimation

Lo et al. [2002] established econometric foundations for execution probability modelling. Maglaras et al. [2022] introduced a deep learning approach treating execution as a time-to-fill distribution. Arroyo et al. [2024] advanced this with a convolutional-transformer using survival analysis, handling the right-censoring inherent in limit order data.

That work uses tick-level message data. Our LSTM follows its survival formulation and fails on snapshot data at a median 18.2-second interval. We do not read this as contradicting those results. The likelier reading is that architectures designed for dense sequential data lose their advantage when the sequence is sparse and irregular, which is a boundary condition worth stating.

2.4 Tabular deep learning

Our central comparison connects to a finding from outside finance. Systematic evaluations report that gradient-boosted trees remain competitive with or superior to neural networks on tabular data, particularly at moderate sample sizes with heterogeneous, weakly correlated features [Grinsztajn et al., 2022, Shwartz-Ziv and Armon, 2022]. Our features are engineered scalars (spreads, imbalances, volumes) computed per snapshot, which is tabular in structure even though the underlying process is sequential. Our result is consistent with that literature and, so far as we are aware, has not previously been demonstrated for LOB fill prediction.

2.5 Reinforcement learning for market making

Spooner et al. [2018] demonstrated RL agents learning market making policies in simulation; Guo et al. [2023] added continuous action spaces; Briola et al. [2021] established viability in cryptocurrency settings. RELAYER [Jiang et al., 2025] models random execution delays of 30–100 ms. We do not extend this line in the present version (Section 7.4).

3 Problem Formalisation

3.1 Standard setup

A market maker operates over horizon $[0, T]$. The mid-price follows $S_t = S_0 + \sigma W_t$ with $\sigma > 0$ and W_t standard Brownian motion. Posting a bid at distance δ^b and ask at δ^a from mid, fills arrive as

Poisson processes with intensities

$$\lambda^b(\delta^b) = Ae^{-k\delta^b}, \quad \lambda^a(\delta^a) = Ae^{-k\delta^a}, \quad (1)$$

where $A, k > 0$ govern decay of arrival rate with distance from mid. The objective is

$$\max_{\delta^b, \delta^a} \mathbb{E}[X_T + q_T S_T - \gamma q_T^2], \quad (2)$$

with $\gamma > 0$ the inventory risk aversion parameter.

3.2 Execution latency

We introduce execution latency ℓ , the delay between quoting decision and order arrival, and consider $\ell \in [200 \text{ ms}, 2000 \text{ ms}]$. A quote posted at t arrives at $t + \ell$ into a book state $\mathcal{F}_{t+\ell}$ unobserved at decision time, so the relevant fill probability is

$$P^{\text{fill}}(\delta, \ell, \mathcal{F}_t) = \mathbb{P}[\text{order fills} \mid \delta, \mathcal{F}_{t+\ell}], \quad (3)$$

which depends on a future state. This produces *adverse selection amplification*, where quotes are picked off as the mid moves during transmission, and *cancellation impotence*, where an adverse signal cannot be acted on before a fill occurs.

3.3 The latency-compensating problem

Let $\hat{P}^{\text{fill}}$ estimate fill probability and $\hat{S}_{t+\ell}$ the mid-price at arrival. The problem is

$$\max_{\delta^b, \delta^a} \mathbb{E}[X_T + q_T \hat{S}_{t+\ell} - \gamma q_T^2 \mid \hat{\mathcal{F}}_{t+\ell}] \quad (4)$$

subject to participation constraints

$$\hat{P}^{\text{fill}}(\delta, \ell, \mathcal{F}_t) \geq \tau_{\text{fill}}, \quad \left| \hat{S}_{t+\ell} - S_t \right| \leq \Delta_{\text{safe}}, \quad (5)$$

with abstention whenever either is violated.

3.4 Viability conditions

Version 1 stated three conditions: sufficient predictive accuracy, adequate uninformed order flow [Chen and Moallemi, 2018], and strict abstention discipline. This paper concerns the first. Our result is that a usable estimator $\hat{P}^{\text{fill}}$ is obtainable at retail data frequency, which establishes the condition is satisfiable. Whether the accuracy suffices for profitability, and whether the other two conditions hold, is untested here.

4 Data

4.1 Collection

LOB data were collected from Kraken via WebSocket API v2, subscribing to the `book` channel at depth 10 for BTC/USD, ETH/USD and SOL/USD. Updates are timestamped at receipt and stored with the full 10-level ladder in Parquet, partitioned by asset, date and hour.

Collection ran from 15 June to 24 August 2026. The collector is managed by a systemd unit with `Restart=always`; over the final three weeks it restarted automatically three times after WebSocket disconnections. An earlier configuration using a detached terminal session and a bounded reconnect loop failed silently on 20 July and stayed down for fourteen days, costing an estimated 90,000 snapshots. We note this because unattended collection reliability is a material and under-discussed constraint on independent microstructure research.

4.2 Sampling characteristics

Kraken’s book channel is event-driven, so sampling is sparse and irregular.

Table 1: Inter-update intervals, BTC/USD.

Statistic	Value
Mean gap	40.1 s
Median gap	18.2 s
Minimum gap	<0.01 s
Maximum gap	6339 s
Updates per hour	≈ 90

The maximum gap corresponds to the collector outage rather than market quiescence. This sampling rate is three to four orders of magnitude coarser than the tick data used in Arroyo et al. [2024] and Maglaras et al. [2022], which is the context in which our LSTM result should be read.

Table 2: Collected dataset, 15 June – 24 August 2026.

Asset	Snapshots	Label rows	Overall fill rate
BTC/USD	111,002	3,551,392	0.734
ETH/USD	34,380	1,099,488	0.754
SOL/USD	123,948	3,965,664	0.534
Total	269,330	8,616,544	—

4.3 Label construction

For each snapshot we simulate limit order submissions at offsets $\delta \in \{1, 2, 3, 5\}$ ticks on each side, recording whether each would have filled within each latency window $\ell \in \{200, 500, 1000, 2000\}$ ms based on subsequent book evolution. This yields $4 \times 4 \times 2 = 32$ label rows per snapshot, with unfilled orders right-censored at ℓ .

Both conditioning variables show strong monotone structure (Table 3), which is what makes the lookup table a demanding reference rather than a straw one.

4.4 Features

From each snapshot we construct 109 features: 40 raw price and volume levels, normalised prices and volumes, mid-price, spread, spread in basis points, order book imbalance, depth imbalance, and rolling statistics and lagged differences. Prices are normalised by contemporaneous mid, volumes by a rolling mean.

Table 3: Marginal fill rates by quote offset and latency window, SOL/USD.

Offset (ticks)	Fill rate	Latency (ms)	Fill rate
1	0.782	200	0.385
2	0.550	500	0.498
3	0.446	1000	0.585
5	0.358	2000	0.669

We define a *microstructure* subset of 63 features by removing every price-derived column, raw and normalised, retaining spreads, imbalances, order book imbalance, depth, volumes and volatility measures. This subset is used for the headline results; the rationale is in Section 8.3.

Two pipeline parameters required adjustment for sparse data. The rolling normalisation window was reduced from 1000 to 10 snapshots and lag features from $\{1, 5, 10, 20, 50\}$ to $\{1, 2, 3, 5\}$. At the original settings a 1000-snapshot window spans roughly eleven hours of wall-clock time, which is not a meaningful local normalisation.

5 Models and Evaluation

5.1 The reference

Our benchmark is a lookup table: the mean fill rate for each $(\delta, \ell, \text{side})$ cell, 32 cells, estimated on training data and applied unchanged to test data. It has no learned parameters, no access to the order book and no temporal information.

It is a demanding reference. Quote offset spans a 0.42 range in fill rate and latency 0.28 (Table 3). Any model failing to beat it adds nothing over two variables the trader selects directly and already knows.

5.2 Gradient-boosted trees

We fit gradient-boosted trees (XGBoost, 400 estimators, depth 6, learning rate 0.05, subsample 0.8) on the microstructure subset for the current snapshot only, plus the three conditioning variables. The model receives **no temporal window**.

5.3 LSTM survival model

The LSTM encodes a lookback of $L = 50$ snapshots with two layers (hidden dimension 64, dropout 0.2). A conditioning vector $c = [\ell/\ell_{\max}, \delta/\delta_{\max}, \text{side}] \in \mathbb{R}^3$ passes through an MLP and is concatenated before a monotone survival decoder. Total parameters: 90,754. Training minimises the negative log-likelihood under right-censoring,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[\Delta_i \log h(\tilde{T}_i) + \log S(\tilde{T}_i) \right], \quad (6)$$

using Adam at $\eta = 10^{-3}$ with early stopping.

The comparison is deliberately unfavourable to the trees. The LSTM sees fifty snapshots of history and 46 additional price columns; the trees see one snapshot and fewer features.

5.4 Splitting

All results use a *snapshot-grouped* temporal split at 70/15/15. Because each snapshot generates 32 label rows sharing identical features, a row-wise split lets a snapshot appear in both training and test data, allowing a sufficiently flexible model to memorise it. Grouping on snapshot removes this. The effect is measured in Section 8.

5.5 Metric

We report fill AUC: area under the ROC curve for the binary filled/not-filled task, computed exactly via the Mann-Whitney rank-sum identity with ties credited 0.5. AUC is the relevant quantity because the abstention rule in (5) tests $\hat{P}^{\text{fill}} \geq \tau_{\text{fill}}$, a binary decision that never compares timings.

6 A Conditioning Defect and Its Use as a Control

Our first real-data results were substantially worse than chance. Our initial hypothesis was insufficient data. That hypothesis was wrong, and the diagnostic that displaced it is worth describing.

6.1 The defect

The conditioning vector c originally contained only ℓ . Quote offset δ and side were used to construct labels but never reached the model.

For a fixed snapshot and latency, the eight label rows spanning four offsets and two sides therefore present *identical* model input while carrying different targets, with marginal fill rates from 0.782 at 1 tick to 0.358 at 5 ticks. No function of the input can separate them, and the loss-minimising response is to predict the conditional mean and ignore the input. Validation loss moved roughly 0.003 over 40 epochs in every run.

This also explains an observation that misled us. Across collection milestones, measured C-index moved 0.5008, 0.5308, 0.5526 as the dataset grew from 21K to 51K to 103K snapshots. We read this as convergence. All three lie within seed variance of chance, and a later run on 269K snapshots returned 0.4749, worse with more data. That is the expected behaviour when additional data supplies additional mutually contradictory input-target pairs.

6.2 Correction and control

Adding δ and side to c raised C-index from 0.4749 to 0.6266 on an otherwise identical configuration, with validation loss movement increasing roughly eightfold.

We retain the defective configuration as an arm, LOB, in which the LOB window is supplied but δ and side are reset to constants. It is a reproduction control: any claim that a result derives from LOB features should come with a demonstration that the corresponding defective configuration performs at chance. Across five seeds it scores 0.5034 ± 0.031 (Table 5).

6.3 Consequence for prior results

All real-data figures in Versions 1.0–2.0 were produced by the defective configuration and should be disregarded, including the C-index progression above and a projection that 300,000 snapshots would be needed for convergence, which was extrapolated from noise.

7 Results

7.1 Main comparison

Table 4 reports all three predictors on identical test rows under the grouped split.

Table 4: Fill AUC by model class, grouped temporal split, identical test rows. GBM uses the 63-feature microstructure subset and the current snapshot only; the LSTM uses all 109 features and a 50-snapshot window.

Asset	Test rows	Lookup	GBM	LSTM	GBM – Lookup	LSTM – Lookup
BTC/USD	29,788	0.6390	0.8128	0.6271	+0.1738	−0.0119
SOL/USD	29,845	0.6819	0.7786	0.6750	+0.0967	−0.0069
ETH/USD	29,798	0.6505	0.6956	0.5341	+0.0450	−0.1164
Mean					+0.1052	−0.0451

Three observations.

The GBM beats the lookup table on every asset, by +0.174, +0.097 and +0.045. Microstructure features carry predictive content for the replay-defined fill outcome at this sampling rate.

The LSTM beats it on none, and on ETH/USD underperforms by 0.116. Pooled across assets it matched the table to +0.0003 AUC.

The effect size varies fourfold across assets and does not track book thickness in any obvious way. SOL/USD has the thinnest book and the steepest offset gradient but a middling gap; ETH/USD has both the fewest snapshots and the smallest gap, which may reflect its 34,380 snapshots rather than any property of the asset. We report the range rather than leading with the mean.

7.2 The LSTM failure

The LSTM receives strictly more information than the GBM: the same microstructure features, plus a 50-snapshot window, plus 46 additional price-derived columns. It performs worse on all three assets.

Its training behaviour is consistent with underfitting rather than overfitting. On ETH it stopped at epoch 13 with validation loss moving 1.7405 to 1.7329; on SOL at epoch 12; on BTC at epoch 32 with the largest movement, 1.6592 to 1.6183, and the best relative result. Longer patience, a larger hidden dimension or a different learning rate schedule might change this, and we did not run that search.

What we can claim is narrower than “deep learning fails here.” A specific architecture, adapted from work on tick-level data and trained with our hyperparameters, does not recover signal that a standard tabular method recovers easily from less information.

7.3 Ablation

For completeness we report the ablation run on the LSTM before the GBM comparison existed, using C-index on a pooled row-wise split.

FULL exceeds COND by 0.085, which taken alone appears to show the LSTM using LOB features productively. It does not. COND is a model trained with its dominant input stream zeroed, a damaged network rather than a fair alternative predictor, and Table 4 shows the LSTM adds nothing over quote geometry. We report this because the ablation was run first and, alone, supported a conclusion the external reference reversed.

Table 5: LSTM ablation, C-index, five seeds per arm, pooled row-wise split.

Arm	Mean	SD	Min	Max
FULL (LOB + conditioning)	0.6101	0.034	0.5670	0.6536
COND (conditioning only)	0.5247	0.041	0.4853	0.5885
LOB (defective control)	0.5034	0.031	0.4678	0.5427

LOB at 0.5034 confirms the defect of Section 6 reproduces to chance.

7.4 Status of the reinforcement learning component

Version 1 proposed a latency-aware PPO agent consuming fill probability estimates, with a reward penalising adverse fills attributable to execution delay. We report no RL results. The agent was trained only on synthetic data, where it did not converge, and it consumed estimates from the defective model, which makes those runs uninterpretable.

Given the results above, an RL agent is now worth revisiting with GBM-derived fill estimates. We regard this as the most promising immediate extension and it is not attempted here.

8 Audit of the Positive Result

A positive result obtained after several negative ones deserves scrutiny. We report the checks we ran, including those that failed to find a problem.

8.1 Splitting

Our initial GBM result used a row-wise temporal split and scored 0.8264 on BTC/USD. Because 32 label rows share each snapshot’s features, this permits a snapshot to straddle train and test. Regrouping on snapshot reduced the score to 0.8127, a loss of 0.014. The effect is real but small, and all results in Section 7 use the grouped split.

8.2 Shuffled-split contrast

A shuffled split, invalid for time series, scored 0.8258 against 0.8264 for the temporal row split. The near-identity suggests the signal is local and stationary rather than trend-dependent. We note this as an observation rather than a check that was passed, since we do not have a clean interpretation of it.

8.3 Price-level features

Feature importances on BTC initially concerned us. The largest contributors were `bid_price_2` (0.143) and `ask_price_2` (0.119), raw prices in the tens of thousands of dollars. A tree splitting on absolute price level in temporally-ordered data is partitioning on time period, which proxies for regime, and that is not microstructure prediction.

Removing every price-derived column, raw and normalised, tests this directly.

Removing price columns *improved* every asset, by +0.008 on BTC, +0.022 on ETH and +0.006 on SOL. Our hypothesis was wrong: the price columns were nuisance variables consuming splits, not a leakage channel. Headline figures use the 63-feature microstructure subset accordingly.

Table 6: Fill AUC gain over the lookup table by feature set, grouped split.

Feature set	LOB features	BTC	SOL	ETH
All	109	+0.1658	+0.0926	+0.0233
No price columns	68	+0.1724	+0.0982	+0.0461
Microstructure only	63	+0.1738	+0.0967	+0.0450
Conditioning only	0	−0.0007	+0.0000	−0.0007

8.4 Validating the comparison

The conditioning-only row of Table 6 fits a GBM on the three conditioning variables alone. It reproduces the lookup table to within 0.0007 on every asset. That is the expected behaviour, and it confirms the comparison measures what we intend.

8.5 What remains unchecked

We have not tested other venues, other feed types or other tree hyperparameters. We have not run repeated seeds for the GBM, so per-asset figures carry unmeasured variance, though the between-asset spread (+0.045 to +0.174) is larger than seed variance plausibly accounts for. We have not established that the signal is economically exploitable.

9 Methodological Lessons

Three errors in this project were caught only by checks that could have been run at the outset, at a combined cost of about two months. Each is cheap to avoid.

Run the trivial reference first. The lookup table took minutes to implement and reversed our conclusion twice: first by showing the corrected LSTM added nothing, then by providing the scale against which the GBM’s gain was measurable. Without it we would have published the LSTM’s 0.6266 C-index as a modest positive result.

An ablation is not a substitute for an external baseline. Our LSTM ablation showed a clean +0.085 separation between FULL and COND and would, presented alone, have supported the claim that the model used LOB features productively. Both arms were below the lookup table. Ablations measure internal dependence, not external validity.

Do not infer a data limitation from one model’s failure. We reached that inference twice, for different reasons, and it is worth separating them.

The first time, a conditioning defect was feeding the model contradictory input-target pairs (Section 6). Its flat C-index looked like a sample-size problem, and we projected that 300,000 snapshots would be needed for convergence. We then collected for six weeks, adding roughly 130,000 snapshots, and the projection was extrapolated from noise the whole time. More data cannot repair a defective input pipeline.

The second time, with the defect corrected, the model matched a lookup table and we concluded the data contained no usable signal. A different model class overturned that in twenty minutes.

Neither episode licenses the opposite rule of blaming the model first. Failures are sometimes genuinely about data, labels, optimisation or implementation. The transferable version is narrower: before concluding that data contain no usable signal, rule out implementation defects and test at least one model with a different inductive bias.

10 Limitations

No economic evaluation. AUC is not profit. We run no backtest, model no transaction costs, and make no claim that the signal survives the bid-ask spread, maker fees, queue position or adverse selection. This is the largest gap between our result and its motivating application.

Simulated fills. Labels come from replaying subsequent book states, not executed orders. Queue position is not modelled, so we assume a quote at a price level fills when that level trades through, which is optimistic for a latecomer.

One venue. Kraken’s depth-10 book channel only.

One deep architecture. We tested an LSTM survival model with one hyperparameter configuration. The convolutional-transformer of Arroyo et al. [2024], which Version 1 proposed, was not re-evaluated after the conditioning fix.

No GBM seed variance. Per-asset GBM figures come from single fits.

ETH sample size. ETH/USD has 34,380 snapshots against 111,002 and 123,948 for the others, and shows both the weakest GBM gain and the worst LSTM failure. Its results should be read as confounded with sample size.

11 Conclusion

We asked whether limit order book microstructure predicts fill probability at the data frequency available to a retail participant, roughly 90 updates per hour from a public exchange feed. For our replay-defined fill outcome, it does. Gradient-boosted trees over microstructure features beat a lookup table on quote offset, latency and side by +0.174, +0.097 and +0.045 AUC across three cryptocurrency pairs, under a snapshot-grouped temporal split, with price-derived features removed, and with a conditioning-only control that reproduces the reference exactly.

An LSTM survival model over the same features, given a fifty-snapshot temporal window the trees never receive, recovers none of it. It matches the trivial reference pooled and underperforms it on all three assets individually.

We set out to test whether prediction can substitute for speed. We can now say the predictive component is available at retail data frequency, which was genuinely in doubt, and that in our experiments recovering it turned on model selection rather than on collecting more data. Whether the resulting edge survives transaction costs, queue position and adverse selection is the next question and the harder one.

Our own route to this result ran through the opposite conclusion, written up in full. The correction came from a twenty-minute reference we had deferred for two months. We report that sequence because the cost of the error was six weeks of data collection aimed at a problem that did not exist.

Data and Code Availability

The complete collection pipeline, feature engineering, models and every diagnostic reported here are available at <https://github.com/azure-cyber/latency-compensating-mm>. The scripts underlying Sections 7 and 8 (`gbm_matched.py`, `gbm_price_test.py`, `gbm_leakage_check.py`, `ablation.py`, `diagnose.py`) are included so both the positive and negative results can be checked directly.

Disclosure of AI Assistance

The author used Claude (Anthropic) extensively throughout this project: for implementation of the data collection pipeline, feature engineering, models and evaluation code; for diagnostic analysis including identification of the conditioning defect in Section 6 and design of the audit in Section 8; and for drafting and revision of this manuscript. The author designed the study, directed the analysis, made all substantive research decisions, verified the results, and is solely responsible for the content and any errors.

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